Python

Sat Mar 7 11:31 PM

MachineLearning

EXPLORER

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OUTLINE

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exercise1.py U

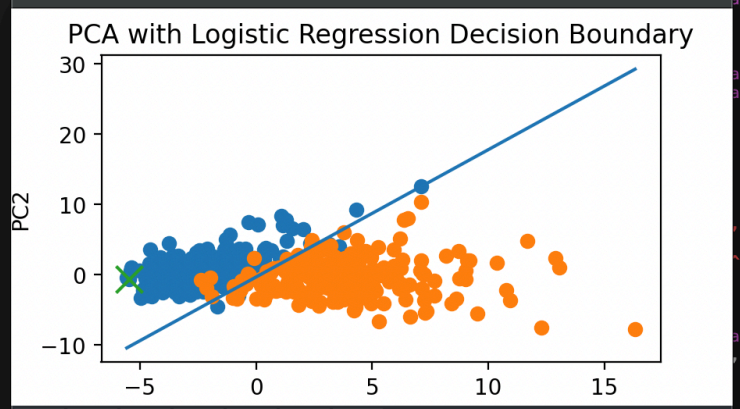
Lab 6 > WilliamsBenjamin > Exercise1 > exercise1.py > ...

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.linear_model import LogisticRegression

df = pd.read_csv("wdbc.data.csv", header=None)

y = df.iloc[:,1].map({"M":1,"B":0}).to_numpy()
X = df.iloc[:,2:].to_numpy(dtype=float)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print("Variance ratio:", pca.explained_variance_ratio_)

model = LogisticRegression()
model.fit(X_pca, y)

dataPoint = np.array([
7.76,24.54,47.92,181,0.05263,0.04362,0,0,0.1587,0.05884,
0.3857,1.428,2.548,19.15,0.007189,0.00466,0,0,0.02676,0.002783,
9.456,30.37,59.16,268.6,0.08996,0.06444,0,0,0.2871,0.07039
])

dataPoint_scaled = scaler.transform(dataPoint.reshape(1,-1))
dataPoint_pca = pca.transform(dataPoint_scaled)

pred = model.predict(dataPoint_pca)[0]
print("Prediction:", pred)

plt.figure()

m0 = y==0
m1 = y==1

plt.scatter(X_pca[m0,0], X_pca[m0,1])
plt.scatter(X_pca[m1,0], X_pca[m1,1])

x_vals = np.linspace(X_pca[:,0].min(), X_pca[:,0].max(), 200)

w0 = model.intercept_[0]
w1, w2 = model.coef_[0]

y_vals = -(w0 + w1*x_vals)/w2

plt.plot(x_vals, y_vals)
```

CHAT TERMINAL

```
ent/MachineLearn
ing/.venv/bin/py
thon "/Users/ben
williams/Documen
ts/Development/M
achineLearning/L
ab 6/WilliamsBen
jamin/Exercise1/
exercise1.py"
Traceback (most recent call last):
  File "/Users/benwilliams/Documents/Development/MachineLearni
ng/Lab 6/WilliamsBenjamin/Exercise1/exercise1.py", line 8, in
<module>
    df = pd.read_csv("wdbc.data.csv", header=None)
  File "/Users/benwilliams/Documents/Development/MachineLearni
ng/.venv/lib/python3.13/site-packages/pandas/io/parsers/reader
s.py", line 873, in read_csv
    return _read(filepath_or_buffer, kwds)
  File "/Users/benwilliams/Documents/Development/MachineLearni
ng/.venv/lib/python3.13/site-packages/pandas/io/parsers/reader
s.py", line 300, in _read
    parser = TextFileReader(filepath_or_buffer, **kwds)
```

Figure 1




```
FileNotFoundError: [Errno 2] No such file or directory: 'wdbc.
data.csv'
benwilliams@Bens-MacBook-Air-2 Exercise3 % cd ../..
benwilliams@Bens-MacBook-Air-2 Lab 5 % cd ../Lab\ 6
benwilliams@Bens-MacBook-Air-2 Lab 6 % cd WilliamsBenjamin
benwilliams@Bens-MacBook-Air-2 WilliamsBenjamin % cd Exercise1
benwilliams@Bens-MacBook-Air-2 Exercise1 % /Users/benwilliams/
Docum
ents/Development/MachineLearning/.venv/bin/python "/Users/benw
illia
ms/Documents/Development/MachineLearning/Lab 6/WilliamsBenjami
n/Exe
rcise1/exercise1.py"
Variance ratio: [0.44272026 0.18971182]
Prediction: 0
```

zsh

Python Exe...

main* 0 0 Quarto: 1.7.33 Ln 13, Col 26 Spaces: 4 UTF-8 LF Python .venv (3.13.7)

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OUTLINE

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Lab 6 > WilliamsBenjamin > Exercise2 > exercise2.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB

df = pd.read_csv("golf.csv")

encoders = {}
for col in df.columns:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    encoders[col] = le

X = df.iloc[:,:-1].to_numpy()
y = df.iloc[:,-1].to_numpy()

model = GaussianNB()
model.fit(X,y)

samples = [
["Rainy","Hot","High","True"],
["Sunny","Mild","Normal","False"],
["Sunny","Cool","High","False"]
]

samples_encoded = []

for row in samples:
    encoded = []
    for i,col in enumerate(df.columns[:-1]):
        encoded.append(encoders[col].transform([row[i]])[0])
    samples_encoded.append(encoded)

samples_encoded = np.array(samples_encoded)

preds = model.predict(samples_encoded)

print("Predictions:")
for s,p in zip(samples,preds):
    print(s,"->", encoders["PlayGolf"].inverse_transform([p])[0])
```

CHAT | TERMINAL

```
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "/Users/benwilliams/Documents/Development/MachineLearning/.venv/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 1645, in __init__
    self._engine = self._make_engine(f, self.engine)
                   ~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^
  File "/Users/benwilliams/Documents/Development/MachineLearning/.venv/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 1904, in _make_engine
    self.handles = get_handle(
                   ~~~~~~~~~~^
        f,
        ^^
    ...<6 lines>...
        storage_options=self.options.get("storage_options", None),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "/Users/benwilliams/Documents/Development/MachineLearning/.venv/lib/python3.13/site-packages/pandas/io/common.py", line 926, in get_handle
    handle = open(
        handle,
    ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'wdbc.data.csv'
benwilliams@Bens-MacBook-Air-2 Exercise3 % cd ../..
benwilliams@Bens-MacBook-Air-2 Lab 5 % cd ../Lab\ 6
benwilliams@Bens-MacBook-Air-2 Lab 6 % cd WilliamsBenjamin
benwilliams@Bens-MacBook-Air-2 WilliamsBenjamin % cd Exercise1
benwilliams@Bens-MacBook-Air-2 Exercise1 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 6/WilliamsBenjamin/Exercise1/exercise1.py"
Variance ratio: [0.44272026 0.18971182]
Prediction: 0
benwilliams@Bens-MacBook-Air-2 Exercise1 % cd ../Exercise2
benwilliams@Bens-MacBook-Air-2 Exercise2 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 6/WilliamsBenjamin/Exercise2/exercise2.py"
Predictions:
Traceback (most recent call last):
  File "/Users/benwilliams/Documents/Development/MachineLearning/Lab 6/WilliamsBenjamin/Exercise2/exercise2.py", line 40, in <module>
    print(s,"->", encoders["Play"].inverse_transform([p])[0])
                  ~~~~~~~~^^^^^^^
KeyError: 'Play'
benwilliams@Bens-MacBook-Air-2 Exercise2 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 6/WilliamsBenjamin/Exercise2/exercise2.py"
Predictions:
['Rainy', 'Hot', 'High', 'True'] -> No
['Sunny', 'Mild', 'Normal', 'False'] -> Yes
['Sunny', 'Cool', 'High', 'False'] -> No
benwilliams@Bens-MacBook-Air-2 Exercise2 %
```